

Arithmetic Expression Evaluator in C++	Version: 1.0
Test Cases for Front End	Date: 2026-05-05

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Revision History

Date	Version	Description	Author
2026-05-05	1.0	Front End Test Cases	Chase Hampton

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Test Cases

1. Purpose

This document defines test cases for the Front-End module of the complete Project. The purpose of these tests is to verify correct uses of the Front-End and that all inputs work from the user to work properly as well as receiving the proper outputs from those inputs. This will also make sure that history is accessible and displayed properly for future use.

2. Front End Button Functionality

2.1 Test case identifier

TC-001

2.2 Test item

Tests the Front-End's ability to make sure that all buttons display properly.

2.3 Input specifications

Press each button to make sure that their use case is displayed properly on screen.

2.4 Output specifications

Result:

2.5 Environmental needs

Standard execution environment.

3. Basic Addition and Subtraction Test

3.1 Test case identifier

TC-002

3.2 Test item

Tests whether the Front end can perform basic addition and subtraction abilities

3.3 Input specifications

Input expression: $2 + 3 - 4$

3.4 Output specifications

Expected Output: 1

Actual Output: 1

Result: Passed

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3.5 Environmental needs

Standard execution environment.

4. Basic Multiplication and Division Test

4.1 Test case identifier

TC-003

4.2 Test item

4.3 Input specifications

Input expression: $3 \times 4 / 2$

4.4 Output specifications

Expected Output: 6

Actual Output: 6

Result: Passed

4.5 Environmental needs

Standard execution environment.

5. Decimal Equation Test

5.1 Test case identifier

TC-004

5.2 Test item

Tests whether the decimal function works properly when implemented into an equation and combined with other decimals.

5.3 Input specifications

Input Expression: $0.05 + 0.10 \times 3$

5.4 Output specifications

Expected Output: 0.45

Actual Output: 0.45

Result: Passed

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5.5 Environmental needs

Standard execution environment.

6. Exponent Equation Test

6.1 Test case identifier

TC-005

6.2 Test item

This tests whether the exponent function will correctly calculate an equation that contains it.

6.3 Input specifications

Input Expression: 3 ** 3

6.4 Output specifications

Expected Output: 27

Actual Output: Nothing

Result: Failed

6.5 Environmental needs

Standard execution environment.

7. Modulo Division Test

7.1 Test case identifier

TC-006

7.2 Test item

Verifies the correctness of modulo division on the calculator.

7.3 Input specifications

Input expression: 4 % 3

7.4 Output specifications

Expected Output: 1

Actual Output: Nothing

Result: Failed

7.5 Environmental needs

Standard execution environment.

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8. Parentheses Equation Test

8.1 Test case identifier

TC-007

8.2 Test item

Tests if the front end can handle equations that contain parentheses.

8.3 Input specifications

Input expression: $(2+3) * 6$

8.4 Output specifications

Expected Output: 30

Actual Output: 15

Result: Fail

8.5 Environmental needs

Standard execution environment.

9. Multiple Decimal Point Test

9.1 Test case identifier

TC-008

9.2 Test item

Test the front end's ability to reject a number that contains multiple decimal points.

9.3 Input specifications

Input expression: $7.00.1 + 6$

9.4 Output specifications

Expected Output: Nothing

Actual Output: Nothing

Result: Passed

9.5 Environmental needs

Standard execution environment.

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10. Division by Zero Test

10.1 Test case identifier

TC-009

10.2 Test item

Tests if the front end will handle a divide by zero problem correctly.

10.3 Input specifications

Input expression: 10/0

10.4 Output specifications

Expected Output: Division by zero error

Actual Output: Error: Division by zero at position 3: cannot divide by zero

Result: Passed

10.5 Environmental needs

Standard execution environment.